

Practical No. 04

ZOO 1203 - General Entomology

Insect Leg, Antennae and Wing Venation

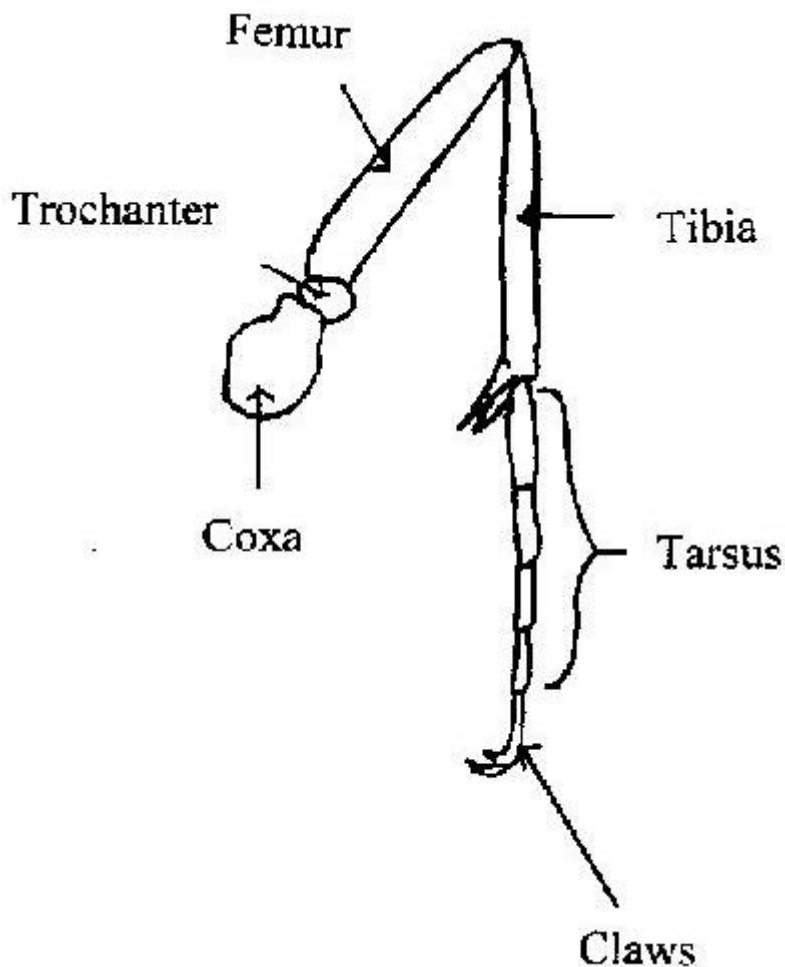
Objectives:

After completing the practical you should be able to;

- Identify different parts of an insect leg
 - Describe the modifications of legs to perform various functions
 - Identify different parts of an insect antennae
 - Identify different types of antennae
 - Get an understanding about wing venation
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1. Insect Leg

1) Examine the different parts of the Meta leg of grasshopper and draw a labeled diagram.



Parts of an insect leg

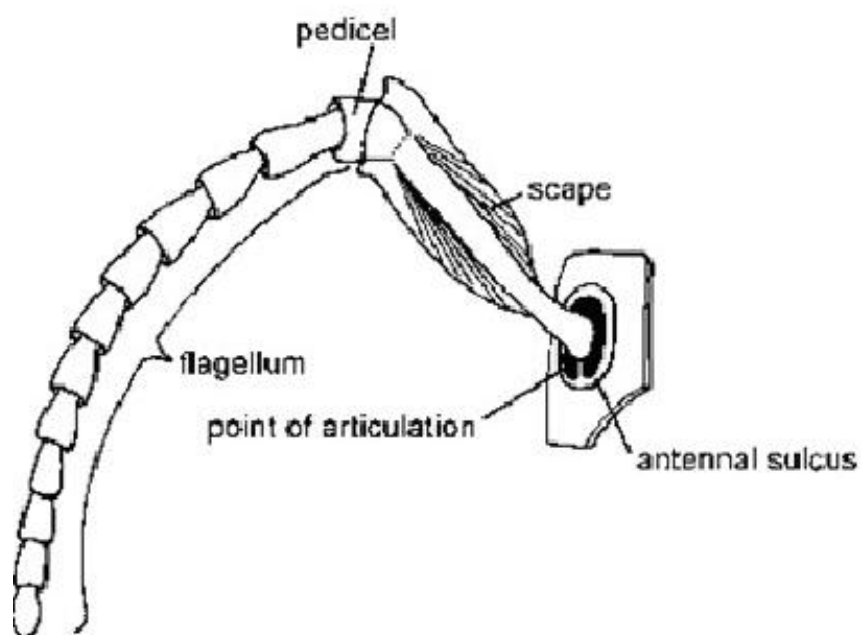
2) Study the modification of legs for performing various functions

- a) Pro leg of mole cricket
- b) Legs of cockroach
- c) Meta leg of grasshopper
- d) Meta leg of worker honey bee
- e) Pro leg of honey bee

2. Different types of Insect Antennae

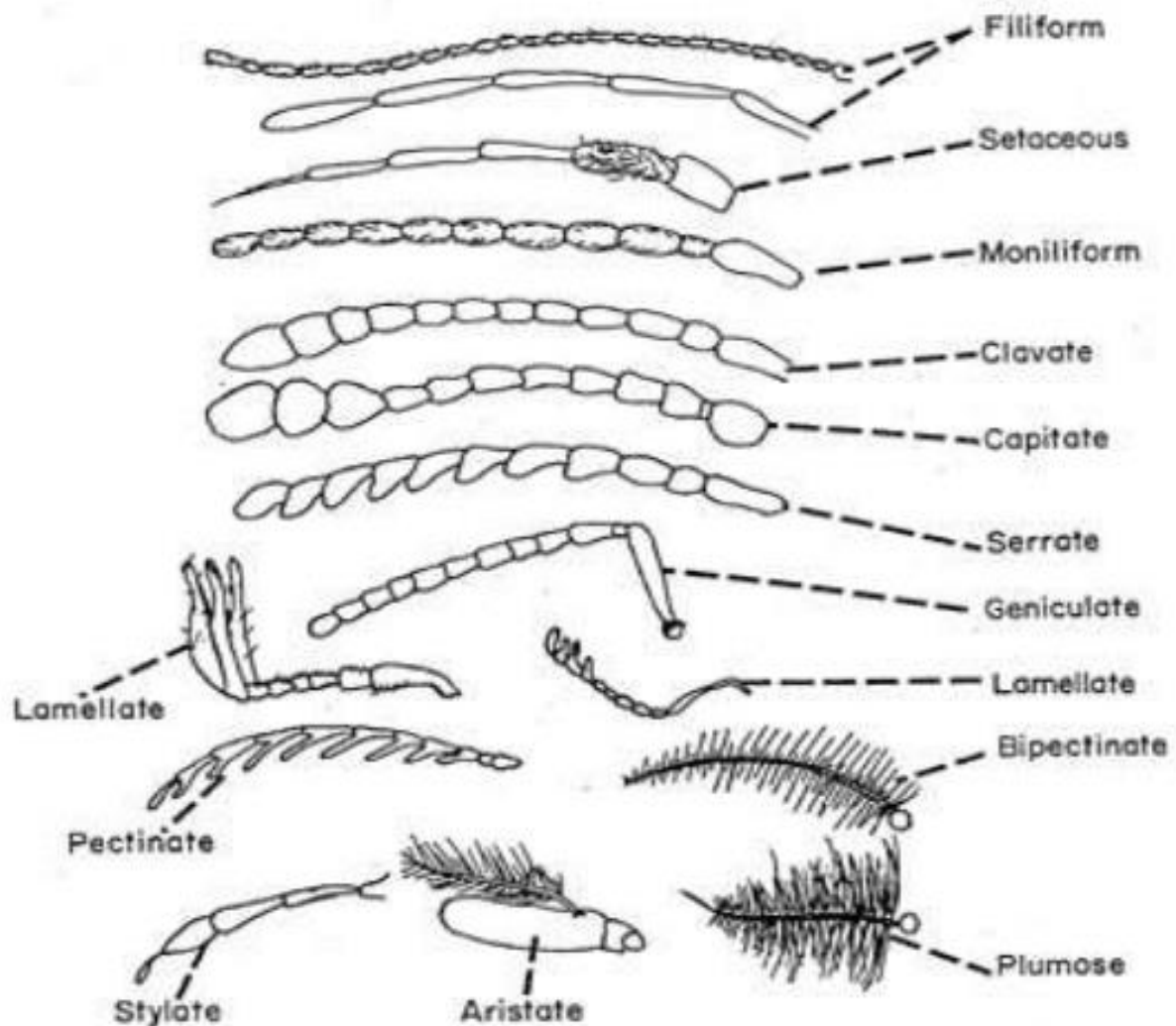
1) Study the different parts of antennae given below.

- a) Scape
- b) Pedicel
- c) Flagellum



Parts of insect antennae

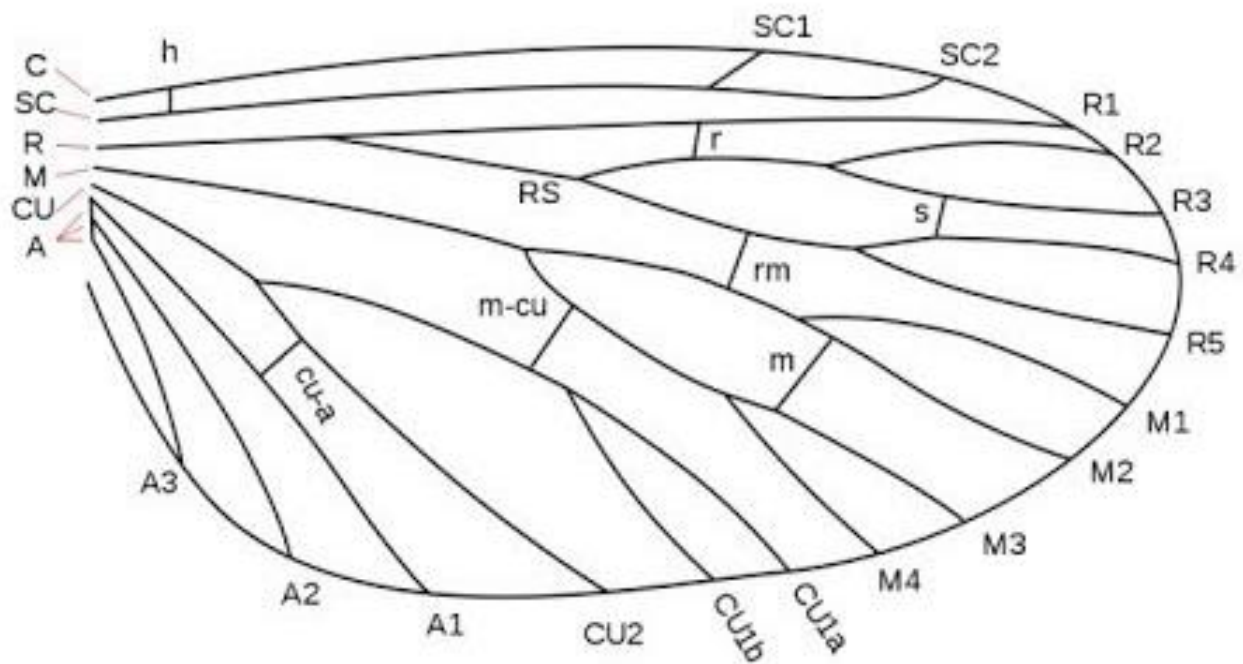
- 2) Examine the insect specimens given to you and study the different types of antennae found on them as depicted in the diagram below.



Modification of insect antennae

Draw the diagrams of various types of antennae you observed on these specimens.

3. Wing Venation



Comstock- Needham system

Wing venation is a defining characteristic of many taxonomic groups. Over the years different names have been ascribed to the pattern of veins, but the most popular and widely used is the Comstock- Needham system.

Costa (C) - the vein along the leading edge of the wing

Subcosta (SC) - second longitudinal vein

Radius (R) - third longitudinal vein

Media (M) - fourth longitudinal vein

Cubitus (CU) - fifth longitudinal vein

Anal (A) - the remaining veins between the cubitus and the trailing edge of the wing

The **cross-veins** connect the longitudinal veins and are typically named for the two veins which they connect.

c-sc - crossvein connecting the costa and subcostal (not on diagram)

r - crossvein connecting adjacent branches of the radius

r-s - crossvein connecting the subcostal and radius (not on diagram)

r-m - crossvein connecting the radius and media

m - crossvein connecting adjacent branches of the media

m-cu - crossvein connecting the media and cubitus

cu-a - crossvein connecting the cubitus and the first anal vein

Certain additional cross-veins have special names.

h- humeral - this cross-vein connects the costa and subcostal near the base of the wing

s- sectorial - a cross-vein that connects either the radius(R1) to the branch R4-R5, or connects R3 to R4

Observe and draw the wing venations of the provided specimens. Try to identify the sections.